South Okanagan Bat Hibernacula Project 2019-2025

2025-26 Update

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1 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on Treaty 8 and Douglas Treaties Territories as well as the traditional and unceded territories of First Nations and Métis Peoples across all regions of British Columbia, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

2 Background

There is growing evidence that the majority of BC bat species overwinter in the province, and several are occasionally active in winter across British Columbia. Winter acoustic activity has been documented across the province in cave and non-cave habitats (BC BatCaver Program, Lausen 2020), at cliff roosts suspected to be hibernacula in the Peace region (WLRS unpublished data), in coastal forests (WLRS unpublished data), and at five parks in southern BC (BC parks unpublished data). There are likely many year-round acoustic monitoring efforts not mentioned here.

While activity has been recorded at diverse habitats such as cliffs, old forests, and waterbodies, little is known about non-cave microsites used by western bats in these habitats in winter (Nagorsen et al. 1993; Weller et al. 2018). Documented non-cave roosts include talus/tree roosts in southeast Alaska (Blejwas et al. 2021), tree cavities in southeastern BC (de Frietas MSc 2023), a potential talus slope in SE BC (Lausen and Hill 2021) and anthropogenic sites including under siding and trim of houses in coastal BC (BC Community Bat Program unpublished data).

Apart from several large hibernacula known from mines in southern BC, the lack of known hibernacula suggests that the majority of southern BC's bats are believed to hibernate singly or in small groups scattered across the landscape. This would be similar to findings from the western US, where *Myotis* bats in winter are found singly (45 % of winter records) or in small groups, with a median group size of 2 bats, while Townsend's Big-eared bats were in slightly larger groups with median 3 (Weller et al. 2018).

In the Okanagan region, multi-year acoustic monitoring by BC Parks (Paterson 2025) has identified the presence of ten species in late fall/early winter and intermittently throughout the winter period, suggesting hibernation in the vicinity of acoustic monitoring stations. The habitats that have been monitored are all rocky outcrops and canyons. While acoustics have confirmed bat presence in these areas, confirming winter roosting/hibernation in these features and identifying the microsites used in these landscapes will increase understanding of hibernation ecology.

Understanding the winter habitat use and ecology of bats can contribute to management decisions around recreational and industrial activities. Understanding how bats use non-cave features, including rock crevices, talus, and trees, for hibernating will also contribute to our understanding of *Pseudogymnoascus destructans* (fungus causing white-nose syndrome in bats) transmission in BC and influence treatment options.

Winter acoustic sampling has occured in the South Okanagan starting in the 2019 until winter 2026 with the purpose of identifying potential locations for winter hibernacula. The multi-year acoustic monitoring confirmed winter bat activity, which has been interpreted as indicative of hibernacula in the immediate area. However, our field surveys suggested that bats might be flying rapidly through the area, using the site as a flight corridor. Calls recorded on the multi-year detector may, therefore, not be as indicative of immediate, local hibernation as was first believed. Because of these results in 2024/25, the team decided to consider an alternative approach to indentifying netting sites.

In 2024/25 work occured at Doctor's Wall as a capture site to identify local hibernacula microhabitats. This work was led by Skurihina and Kellner, to assess the feasibility of fall/winter bat capture and tracking in South Okanagan parks and protected areas and, to identify microsites used by tagged bats through radiotelemetry. Specifically, this pilot project aimed to

1. Assess the ability to capture bats in late fall/early winter in the south Okanagan, particularly Skaha Park and Haynes Lease Ecological Reserve,
2. begin to identify microsites used by bats in late fall/winter, and
3. test the assumption made in acoustic monitoring that late fall/early winter detections are indicative of bats hibernating in the vicinity – relocation via telemetry would confirm the use of the immediate area.

In 20256/26, we also invested in data capture and sharing, with submission of all acoustic data from long-term monitoring sites to the North American Bat Monitoring Program database and to WildTrax.This will allow contribution to additional larger scale analyses, and storage and re-analysis of acoustic files in future if needed.

Acoustic data from short-term monitoring sites was not submitted to databases. This data and the longer-term monitoring dataset was submitted to BC Wildlife Species Inventory database.

Finally, we invested in developing a data visualization tool that future results can be added into. While initially simple, this tool should be easy to enhance to make it a useful interactive tool to examine patterns in winter bat detections in Okanagan Parks and Protected Areas.

3 Methodology

3.1 Acoustic Monitoring

Passive acoustic monitoring was conducted using Wildlife Acoustic Song Meter MiniBat units, placed out for one to multiple nights, to confirm bat presence and aid in selecting suitable netting microsites. Occasional active monitoring was done during mist-netting, using a handheld Wildlife Acoustic Echometer Touch Pro to monitor bat activity at each capture site. Monitoring effort was not recorded for active monitoring (Table 1). For the 2012 Sarell project acoustic monitoring was done with SM2 Bat and SM2Bat+ units from wildlife acoustics. Most sites were sampled as full-spectrum wave files but the Fortress, Wave, No Climbs and Great White site sin 2012 were sampled only with zero crossing.

Table 1: Parks/ Protected Areas, winter acoustic monitoring sites, and years monitored. LT = Long-term monitoring (duration of months); ST = short-term monitoring (duration of days to weeks).

Site	Latitude	Longitude	Years monitored							Data type	Project report/data source (BC WSI project ID 6430)
			2011-12	2018-19	2019-20	2020-21	2021-22	2022-23	2024-25		
Tplaqin / Enderby Cliffs Park											
Enderby Cliffs	50.5900	-119.0833			x					LT	Paterson 2024[1]
Kalamalka Lake park											
Cougar Canyon	50.1806	-119.2751		x		x	x			LT	Paterson 2024[1]
Okanagan Mountain Park											
Wildhorse Canyon	49.7450	-119.6838			x					LT	Paterson 2024[1]
Okanagan Mountain	49.6741	-119.6092		x		x	x			LT	Paterson 2024[1]
Haynes Lease Ecological Reserve											
Haynes Lease 11	49.0940	-119.5226		x						LT	Paterson 2024[1]
Haynes Lease 21	49.0923	-119.5202			x					LT	Paterson 2024[1]
Haynes Lease 31	49.0930	-119.5179				x	x	x		LT	Paterson 2024[1]
Skaha Bluffs Park											
Fern Gully	49.4501	-119.5617	x							ST	Sarrell 2012[2]
Fortress	49.4371	-119.5676	x							ST	Sarrell 2012[2]
Wave	49.4385	-119.5653	x							ST	Sarrell 2012[2]
No Climbs	49.4390	-119.5627	x							ST	Sarrell 2012[2]
Great White	49.4415	-119.5620	x							ST	Sarrell 2012[2]
Whitewash	49.4348	-119.5666	x							ST	Sarrell 2012[2]
Unknown	49.4357	-119.5627	x							ST	Sarrell 2012[2]
Blazing	49.4403	-119.5631	x							ST	Sarrell 2012[2]
Go Anywhere	49.4434	-119.5620	x							ST	Sarrell 2012[2]
Cave Hill	49.4491	-119.5612	x							ST	Sarrell 2012[2]
Skaha - Great White	49.4424	-119.5619			x					LT	Paterson 2024[1]
Skaha - Fern Gully / Doctors Wall	49.4434	-119.5656		x		x	x	x		LT	Paterson 2024[1]
Gillies Creek Road Detector	49.4301	-119.5622							x	ST	Skurikhina and Kellner 2025[3]
Gillies Creek Open Forest Detector	49.4312	-119.5619							x	ST	Skurikhina and Kellner 2025[3]
Smythe Drive Spring Detector	49.4284	-119.5668							x	ST	Skurikhina and Kellner 2025[3]

[1]Paterson. 2024. Okanagan winter bat monitoring: 5 year summary 2018-2023. Prepared for Kirk Safford, BC Parks.
[2]Sarell. 2012. Late Winter Activity of Bats At Skaha Bluffs. Prepared for: BC Parks Penticton, BC
[3]Skurikhina and Kellner. 2025. South Okanagan Bat Hibernacula Microsite Pilot Project. 2024-25 field update.

Acoustic results for the sites sampled in 2012 were analyzed using Sonobat (version 3.05 Great Basin) if they were full spectrum wave files, and if they were zero-cross files thery were manually reviewd using Analook software. For all other projects data were analyzed using Kaleidoscope Pro (Wildlife Acoustics, version 5.6.8) with bat auto ID Bats of North America 5.8.1 modified for the region. The species list included Pallid Bats (*Antrozous pallidus*, ANPA), Townsend's Big-eared Bat (*Corynorhinus townsendii*, COTO), Big Brown Bat (*Eptesicus fuscus*, EPFU), Hoary Bat (*Lasiurus cinereus*, LACI), Silver-haired Bat (*Lasionycteris noctivagans*, LANO), Californian Myotis (*Myotis californicus*, MYCA), Western small-footed Myotis (*M. ciliolabrum*, MYCI), Long-eared Myotis (*M. evotis*, MYEV), Little Brown Myotis (*M. lucifugus*, MYLU), Fringed Myotis (*M. thysanodes*, MYTH), Long-legged Myotis (*M. volans*, MYVO), and Yuma Myotis (*M. yumanensis*, MYYU). All files were manually reviewed by Brian Paterson and Mandy Kellner.

4 Results

Acoustic sampling happened in a total of 8 years, sampling a total of 20 unique sites. Across all survey years, Cougar Canyon had the highest number of recordings in a single year (2021), with a total of 13,428 detections. The greatest sampling effort in terms of site coverage occurred in , during which 10 unique sites were monitored.

In total there have been 107978 bat recordings recorded over the length of the study. A total of 15 bat species were confirmed across all sites and years. The most frequently recorded species was LANO, with 20,051 detections.

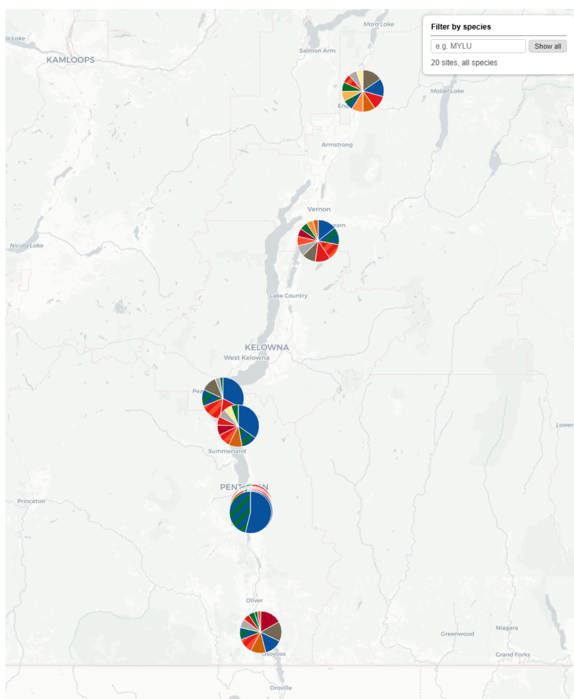

Figure 1: Proportion of species bat passes at each site. Only Species or species couplets that identified at lesat 5% of the total tags appear on map

4.1 Data Submission

The long term acoustic monitoring data set has been formatted and submitted to BC Wildlife Species Inventory under project 6430 - South Okanagan Bat Hibernacula Project. Data has been also been submitted to NABat and can be viewed in the Partner Portal) under project: South Okanagan Bat Winter Bat Project (Project ID:16855). Analyzed acoustic files have been submitted to WildTrax and are available can be found under the project: South Okanagan Bat Winter Bat. Access to data sources will require permission from Mandy Kellner, and can be requested directly through the portals.

5 Discussion

There are several underground cave/talus-cave systems known to BC Parks. Assessment of bat use, species presence, and timing of use could be accomplished through acoustic monitoring and/or guano collection. One such area is at Doctor's wall; during this survey, nets were set up near the entrances of identified caves, but no bats were detected during emergence or caught in the nets. One of the project partners set up an experimental acoustic detector inside the larger cave on November 12, 2024. As of March 2025, this detector had not been retrieved. Options for future work include additional placement of bat detectors at openings or inside caves, single-species DNA analysis through collection of single pellets of guano in paper envelopes (Wildlife Genetics International), and/or multi-species DNA analysis through collection of multiple guano samples (Northern Arizona University Species from Feces lab).

5.1 Recommendations

1. Explore use of detector array to increase understanding of bat winter movements and potential travel corridors as well as inform capture efforts.
2. Target Gilles Creek and the Spring for additional monitoring and capture work as the area currently believed to offer the best chance of successful captures.
3. Increase understanding of bat use of cave/talus features that are known or suspected to be used by bats, through passive acoustic monitoring and guano collection to determine timing of use and species.
4. Expand collection of existing data. This project aims to identify and include additional acoustic datasets. Possible sources including data from 2025/26 Skaha Park winter monitoring (Kellner pers comm), and ECCC/CWS year-round monitoring at Vaseaux (T. Luszcz pers comm)

6 Acknowledgements

This work was supported by funding from the BC Parks License Plate Program, as well as in-kind staff time, vehicles, and equipment from BC Parks and WLRS.

Data included in this compilation and summary was from work done for BC Parks by Brian Paterson and BC Water, Land and Resource Stewardship. Reports and summarized data are available on the BC Wildlife Species Inventory website.

Literature Cited

Loeb, Susan C., Thomas J. Rodhouse, Laura E. Ellison, et al. 2015. *A Plan for the North American Bat Monitoring Program (NABat)*. SRS-GTR-208. U.S. Department of Agriculture, Forest Service, Southern Research Station. https://doi.org/10.2737/SRS-GTR-208.

Olson, Cory. n.d. "Bat Profiles." Alberta Community Bat Program. Accessed March 6, 2025. https://www.albertabats.ca/batprofiles/.

Reichert, Brian, Cori Lausen, Susan Loeb, et al. 2018. *A Guide to Processing Bat Acoustic Data for the North American Bat Monitoring Program (NABat)*. Open-File Nos. 2018–1068. U.S. Geological Survey.

Slough, Brian, Cori Lausen, Brian Paterson, et al. 2022. "New Records about the Diversity, Distribution, and Seasonal Activity Patterns by Bats in Yukon and Northwestern British Columbia." *Northwestern Naturalist* 103 (August): 162–82. https://doi.org/10.1898/NWN21-10.

Solick, Donald. 2022. "Bat Acoustic Species-Pair Matrix Western US/Canada." https://www.batacousticsurveys.com/_files/ugd/a1e0ca_0ed9c3172b9145e49c50eb7a7712f909.pdf.

Szewczak, Joe. 2018. *Acoustic Features of Western US Bats*. Humboldt State University Bat Lab.